```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
df = pd.read_csv("/content/sample_data/Iris.csv")
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: New interactive sheet

```
df = df.drop(['Id'], axis=1)
```

```
X = df.drop(['Species'], axis=1)
```

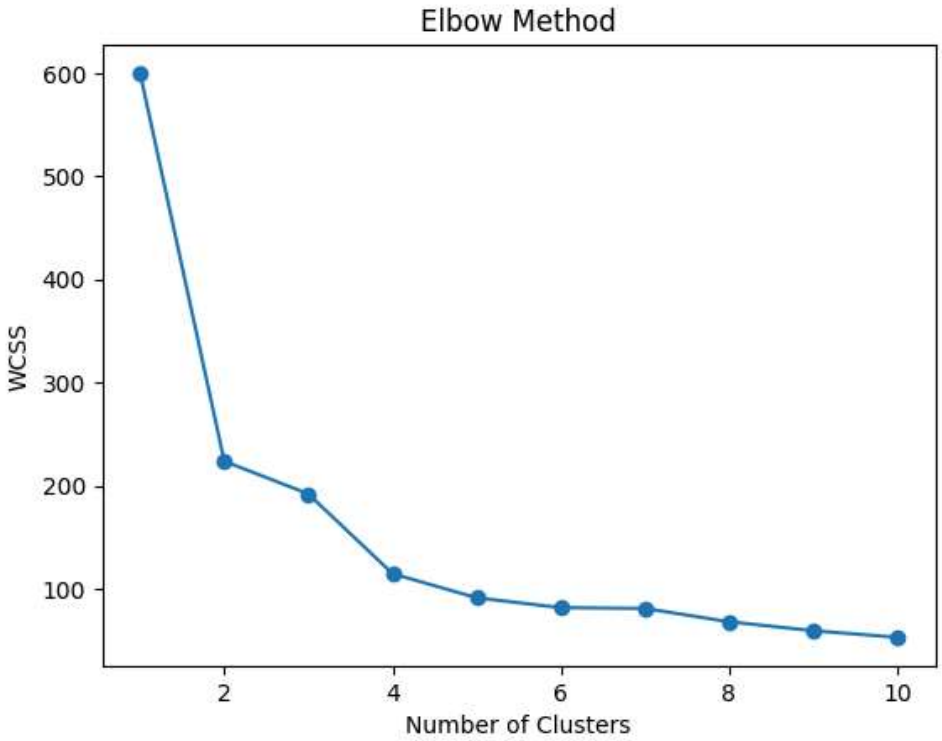
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
wcss = []  # Within Cluster Sum of Squares

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
```

```
plt.figure()
plt.plot(range(1, 11), wcss, marker='o')
```

```
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```



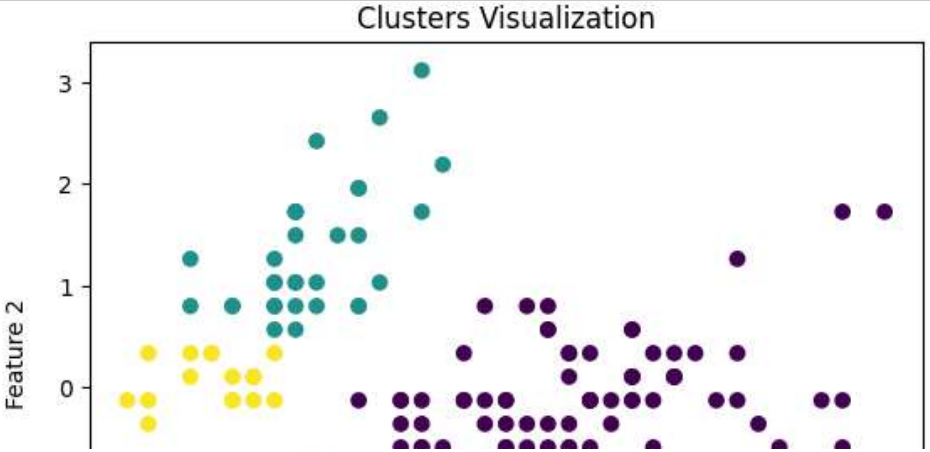

```
kmeans = KMeans(n_clusters=3, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
```

```
df['Cluster'] = y_kmeans
df.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Cluster
0	5.1	3.5	1.4	0.2	Iris-setosa	1
1	4.9	3.0	1.4	0.2	Iris-setosa	2
2	4.7	3.2	1.3	0.2	Iris-setosa	2
3	4.6	3.1	1.5	0.2	Iris-setosa	2
4	5.0	3.6	1.4	0.2	Iris-setosa	1

Next steps: New interactive sheet

```
plt.figure()
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_kmeans)
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('Clusters Visualization')
plt.show()
```


Clusters Visualization
Feature 2
3
2
1
0